

Measurement Conversion, Volume, and Angle Measure

Answer Key

Grades 4–5

Quick Answers

1. 48	4. 82	7. 56	10. 50
2. 3000	5. 2250	8. 145	11. 30
3. 120	6. 5	9. 60	12. 3

Common wrong answers

7. *If they got 146: subtracted from 180 although the two angles form a right angle*
10. *If they got 130: kept the outer-scale reading instead of correcting it*
11. *If they got 30000: reported the volume in cubic centimetres without converting to litres*
12. *If they got 36: multiplied 2 by 18 by 1 without first converting the width from inches to feet*
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1. Jump rope: 4 ft into inches.

Step 1. Feet are the bigger unit, so the count of inches must be larger — multiply, do not divide: $1 \text{ ft} = 12 \text{ in.}$

Step 2. $4 \cdot 12 = 48$.

48 in

2. Bottle: 3 L into millilitres.

Step 1. Litres are the bigger unit, and $1 \text{ L} = 1000 \text{ mL}$, so multiply by 1000.

Step 2. $3 \cdot 1000 = 3000$.

3000 mL

3. Gift box: 8 cm by 5 cm by 3 cm.

Step 1. Volume of a rectangular prism = length $\cdot$ width $\cdot$ height; all three edges are already in centimetres, so no conversion is needed.

Step 2. $8 \cdot 5 = 40$ (the bottom layer holds 40 cubes), and $40 \cdot 3 = 120$ for three layers.

120 cm³

4. $\angle ABD = 35^\circ$, $\angle DBC = 47^\circ$.

Step 1. Ray BD is between the other two rays, so the two small angles add to the whole angle: $\angle ABC = \angle ABD + \angle DBC$.

Step 2. $35 + 47 = 82$.

82°

5. Flour: 2 kg 250 g into grams.

Step 1. Convert the kilogram part first, then add the leftover grams: $1 \text{ kg} = 1000 \text{ g}$.

Step 2. $2 \cdot 1000 = 2000 \text{ g}$, and $2000 + 250 = 2250$.

2250 g

6. Bin: volume 240 cm^3 , base 8 cm by 6 cm.

Step 1. Volume = base area $\cdot$ height, and here the volume is known while the height is missing, so divide instead of multiply.

Step 2. Base area = $8 \cdot 6 = 48$ square cm.

Step 3. $240 \div 48 = 5$, and the check multiplies back: $48 \cdot 5 = 240$.

5 cm

7. Right angle PQR split by ray QS ; $\angle PQS = 34^\circ$.

Step 1. The two parts make a *right* angle, so they add to 90° — not 180° .

Step 2. $90 - 34 = 56$, and the check adds back: $34 + 56 = 90$.

 56° **8. Film: 2 h 25 min into minutes.**

Step 1. Convert the hours, then add the leftover minutes: 1 h = 60 min.

Step 2. $2 \cdot 60 = 120$ min, and $120 + 25 = 145$.

145 min

9. Carton 12 cm by 10 cm by 4 cm packed with 2 cm cubes.

Step 1. Both solids are boxes, so compare volumes: how many cube-volumes fit inside the carton volume?

Step 2. Carton: $12 \cdot 10 \cdot 4 = 480$ cubic cm. Cube: $2 \cdot 2 \cdot 2 = 8$ cubic cm.

Step 3. $480 \div 8 = 60$ cubes. Counting by rows agrees: 6 cubes along the length, 5 across the width, 2 layers high, and $6 \cdot 5 \cdot 2 = 60$.

60

10. Rosa's protractor reading of 130° .

Step 1. A protractor carries two scales running in opposite directions, and the two numbers at any mark always add to 180. Rosa lined her ray up with the zero of one scale but read the number printed on the other.

Step 2. The benchmark check: an acute angle is smaller than a square corner, so its measure must be under 90° . A reading of 130° is more than 90° and contradicts the picture — that is the signal to re-read the other scale.

Model explanation (accept any equivalent wording): “Rosa read the wrong scale. The angle is acute, so the answer has to be under 90° ; I take the other scale's number instead. Before writing any measure down I decide first whether the angle is acute or obtuse, then pick the scale that agrees.”

Step 3. The matching number on the other scale is $180 - 130 = 50$.

 50° **11. Tank 40 cm by 25 cm by 30 cm, answer in litres.**

Step 1. The question asks for litres but the edges are in centimetres, so find the volume in cubic centimetres first and convert at the end.

Step 2. $40 \cdot 25 = 1000$, and $1000 \cdot 30 = 30000$ cubic cm.

Step 3. $30000 \div 1000 = 30$.

30 L

12. Crate 2 ft by 18 in by 1 ft, answer in cubic feet.

Step 1. The three edges are not in the same unit, and a volume in cubic feet needs every edge in feet — convert the width before multiplying.

Step 2. $18 \div 12 = 1.5$, so the width is 1.5 ft.

Step 3. $2 \cdot 1.5 = 3$, and $3 \cdot 1 = 3$. Multiplying $2 \cdot 18 \cdot 1$ instead mixes feet with inches and lands on 36, a number in no unit at all.

 3 ft^3